

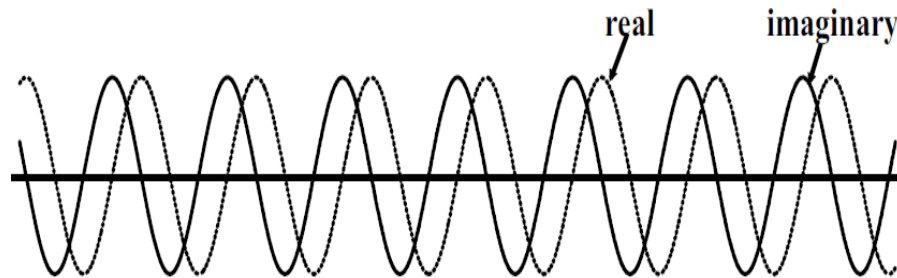
ACML100

Lecture 14

Wave-particle duality and Heisenberg Uncertainty Principle

Can the position be specified exactly ?

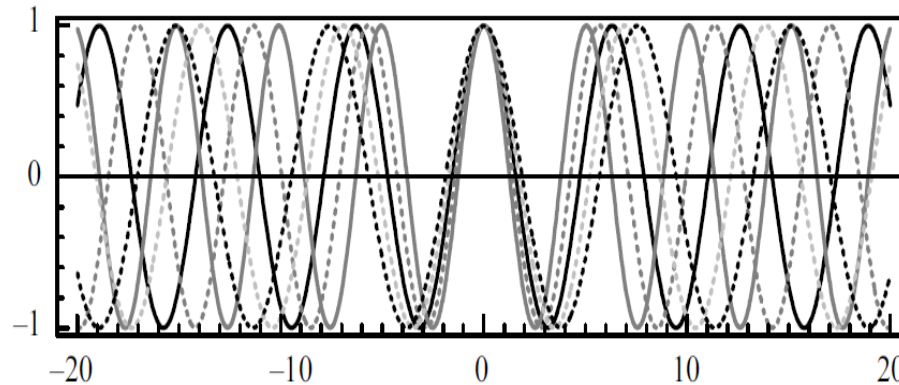
Wavefunction for a free particle with momentum p



Adapted from:
Absolutely small.....,
By Michael D Fayer

- It can have real and imaginary part, having same wavelength.
- They are shifted by one-fourth of wavelength, which is equal to 90 shift in the phase
- They do not interfere either constructively or destructively from each other.
- For this particle with well defined value of momentum, The wavefunction extends from positive infinity to negative infinity.

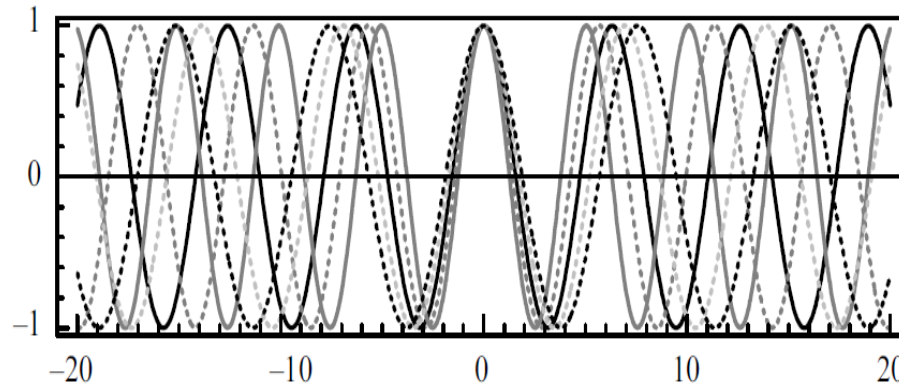
Five Waves of Different Wavelengths



Adapted from:
Absolutely small.....,
By Michael D Fayer

- Five waves of wavelength 1.2, 1.1, 1.0, 0.9, 0.8 are shown.
- The phases are adjusted such that all of the peaks of waves match at 0 on the horizontal axes.
- Because of different wavelengths, they do not match up at other positions.
- At some places, one wave has a positive peak while other has negative peak.

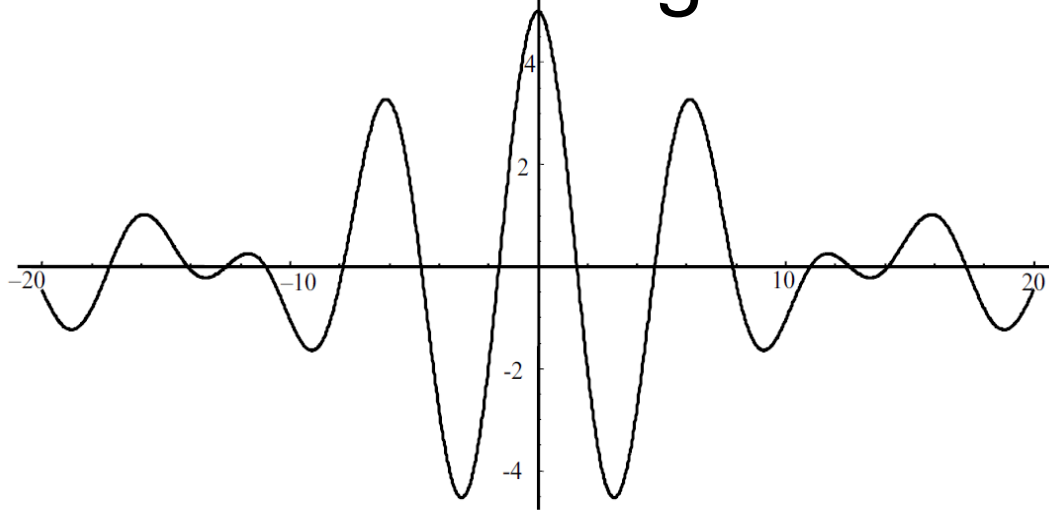
Superposition of Five Waves of Different Wavelengths



Adapted from:
Absolutely small.....,
By Michael D Fayer

- At $x=0$, all of the waves are in same phase, so they add constructively.
- Near $x=0$, the waves are still pretty much in phase.
- In the regions between 10 and 20 and -10 and -20, the difference in wavelengths makes some of the waves positive, where other are negative. So, there is significant cancellation.

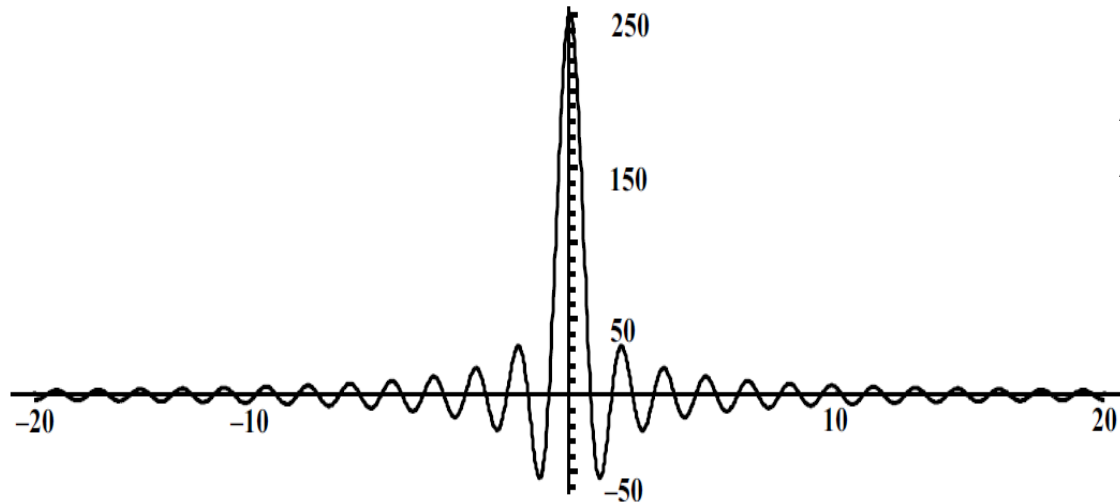
Superposition of Five Waves of Different Wavelengths



Adapted from:
Absolutely small.....,
By Michael D Fayer

- At $x=0$, all of the waves add up constructively, so there is a maxima.
- Next maxima is about $x=6$ and -6 are not as large as maxima at $x=0$.
- In the regions between 10 and 20 and -10 and -20, there is a significant cancellation.

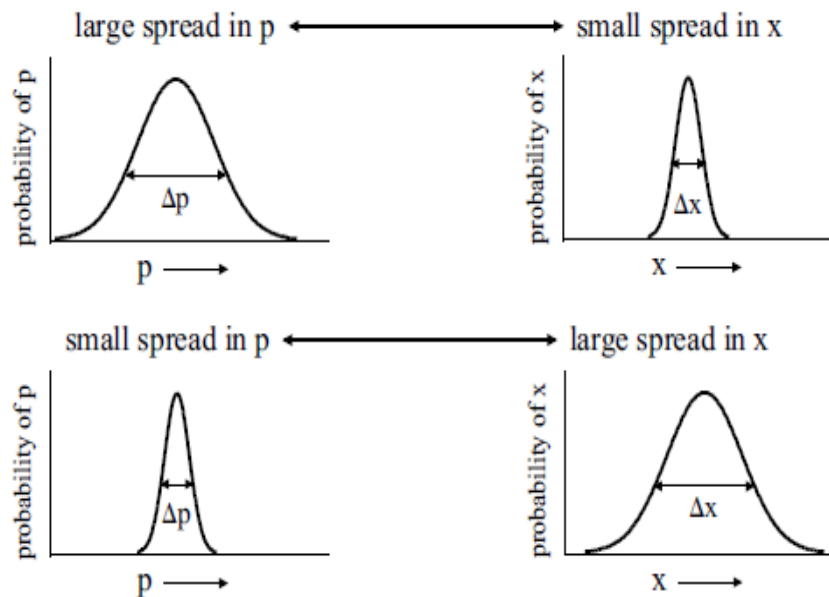
Superposition of 250 Waves of Different Wavelengths between 0 and 4



Adapted from:
Absolutely small.....,
By Michael D Fayer

- Much Larger Maxima at $x=0$, the region of maximum constructive interference.
- Much larger Cancelling at other position of x
- The amplitude of superposition is dying out going towards $+20$ or -20

Relationship between spread in momentum and spread in x



Adapted from:
Absolutely small.....,
By Michael D Fayer

Heisenberg Uncertainty Principle (HUP)

$$\Delta p \Delta x \geq \frac{h}{2\pi}$$

- It is impossible to specify both the momentum and position of the particle simultaneously, with arbitrary precision.
- Higher the uncertainty in momentum, It is easier to locate the particle and vice-versa.

Generalized form of Uncertainty Principle

For any two pairs of observables, the uncertainties in simultaneous determination is related by

$$\Delta O_1 \Delta O_2 \geq \frac{1}{2} \langle [\hat{O}_1, \hat{O}_2] \rangle$$

$$[\hat{O}_1, \hat{O}_2] = \hat{O}_1 \hat{O}_2 - \hat{O}_2 \hat{O}_1$$

UV-Visible spectroscopy

- Absorption of light in the UV/Visible part of the spectrum (210 – 900 nm).
- The transitions that result in the absorption of electromagnetic radiation in this region of the spectrum are transitions between electronic energy levels.
- Generally, the most probable transition is from highest occupied molecular orbital (HOMO) to lowest occupied molecular orbital (LUMO).

A molecule in a one-dimensional box

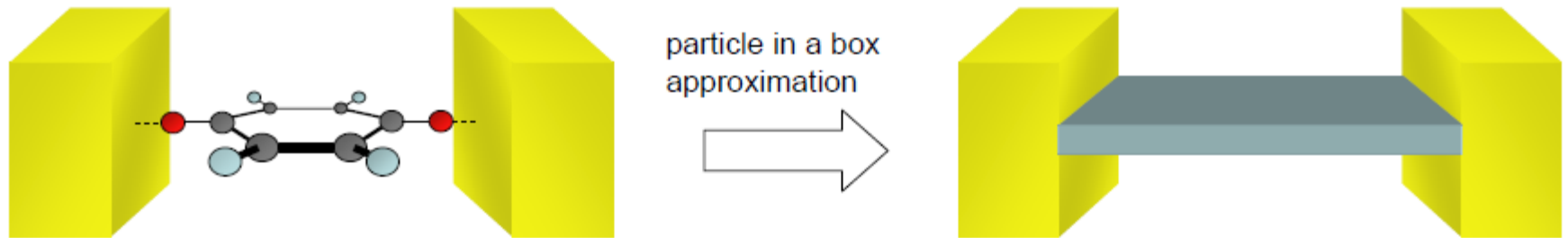
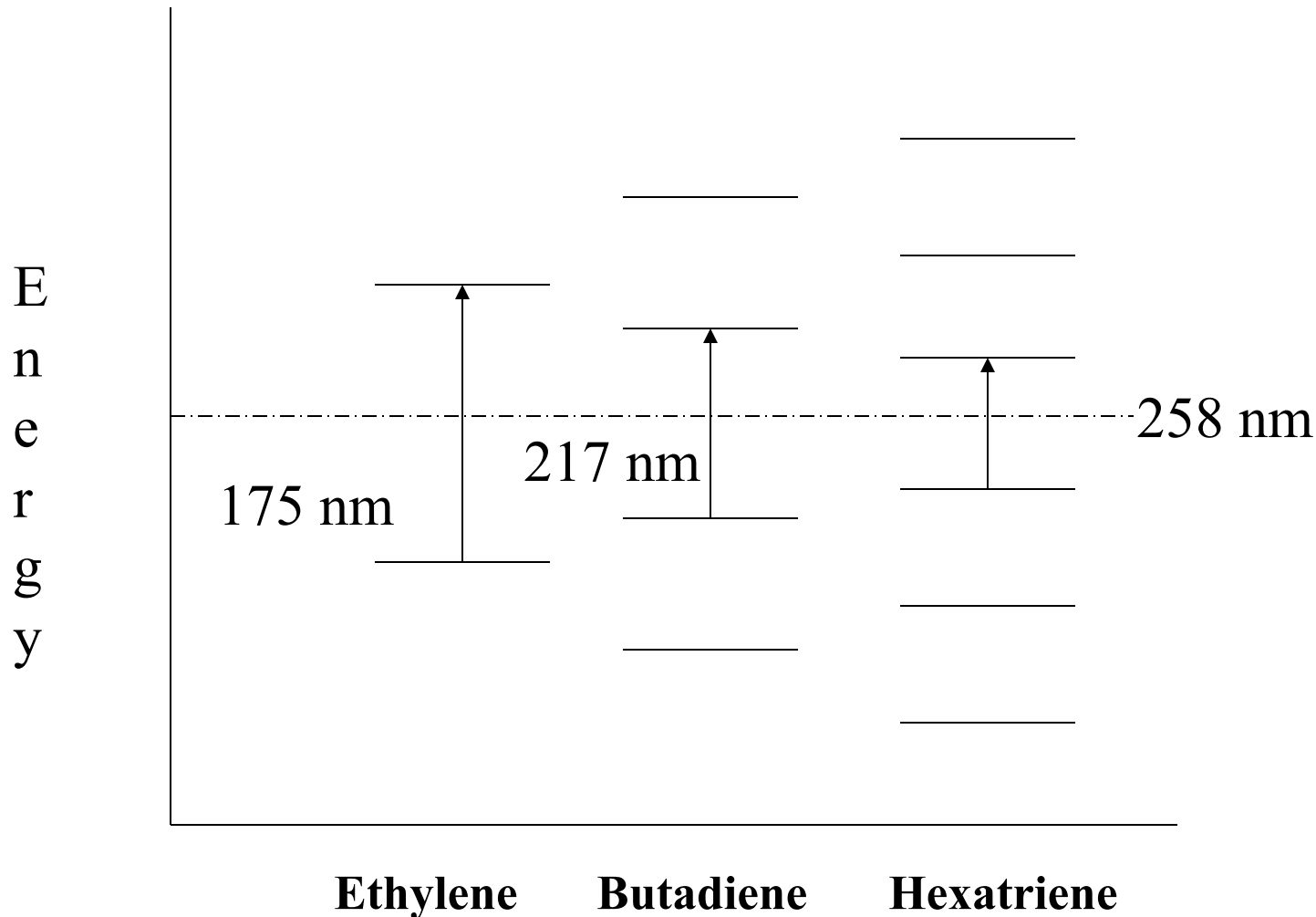


Fig. 2.10. The 'particle in a box' takes a complex structure like a molecule and approximates it by a homogeneous box. All details, such as atoms, are ignored.

The molecules like butadiene can be treated as particle confined in a one-dimensional box.

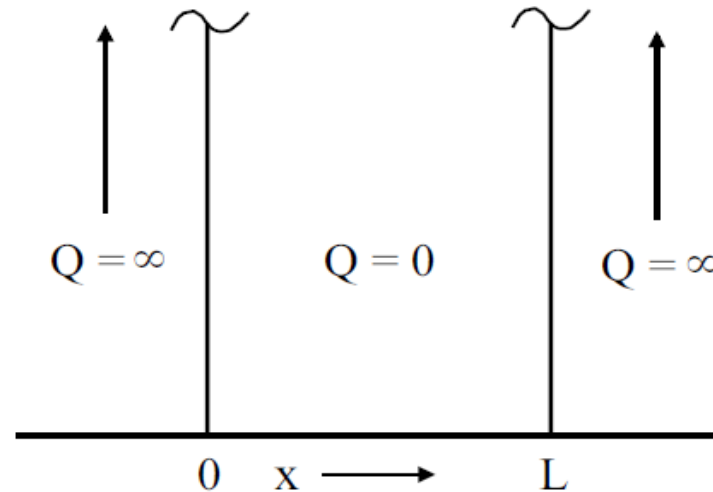
The effect of conjugation of alkene



The Particle in A Box

The 'Classical' Case: A Free particle in a Perfect one-dimensional box

Free
Particle:
No force
acting on it



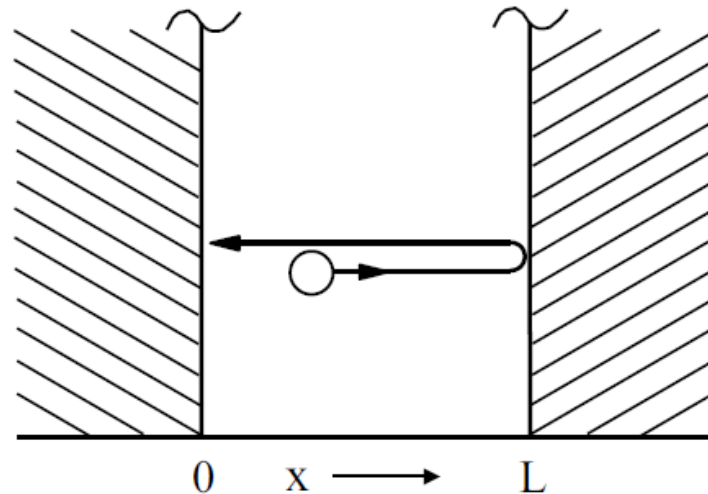
$Q =$
Potential
Energy

Walls are infinitely thick, infinitely massive and completely impenetrable.

Ball in perfect one-dimensional Racquetball court

$$Q=0$$

$$E=K.E$$



L is 12 m and mass of the ball is 0.04 m

If velocity is zero, K.E. is zero, we can know the position. It will be between 0 and L

$$\Delta x=0, \Delta p=0$$

The 'Quantum' Case

Length is 1 nm rather than 12m

Particle has the mass of electron 9.1×10^{-31} kg rather than 0.04 m .

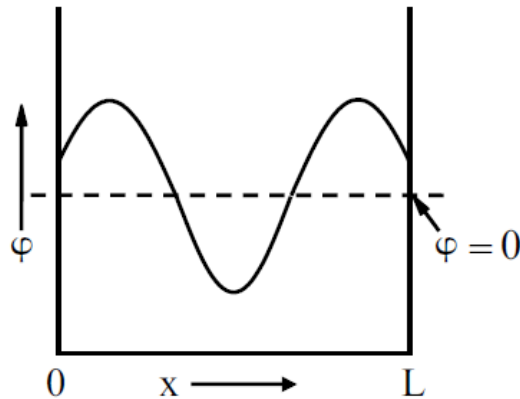
It must obey the Uncertainty Principle

v cannot be zero. The particle cannot be standing still at a specific point.

If v cannot be zero, then E_k can never be zero.

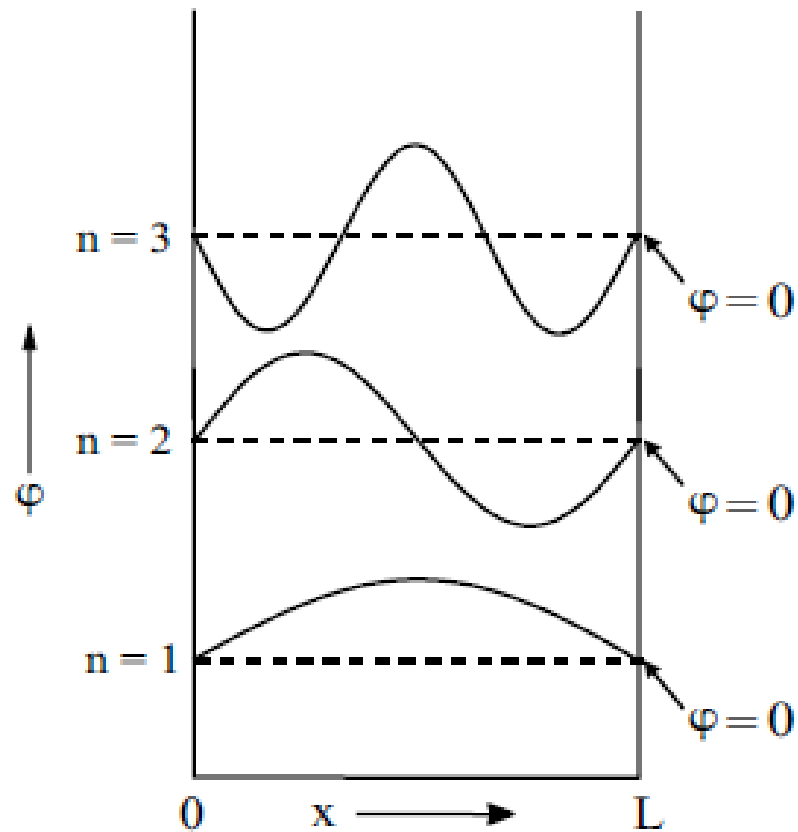
Our quantum racquetball can never stand still.

Energies of a Quantum Particle in a Box



The wavefunction has a nonzero value at the walls and must drop discontinuously to zero outside the box.

Wave functions must be zero at the walls



$$\lambda = 2L/n$$

Energies are quantized

$$E = \frac{1}{2}mV^2$$

$$p^2 = m^2V^2$$

$$E = \frac{p^2}{2m}$$

$$E = \frac{h^2}{2m\lambda^2}$$

$$\lambda = 2L/n$$

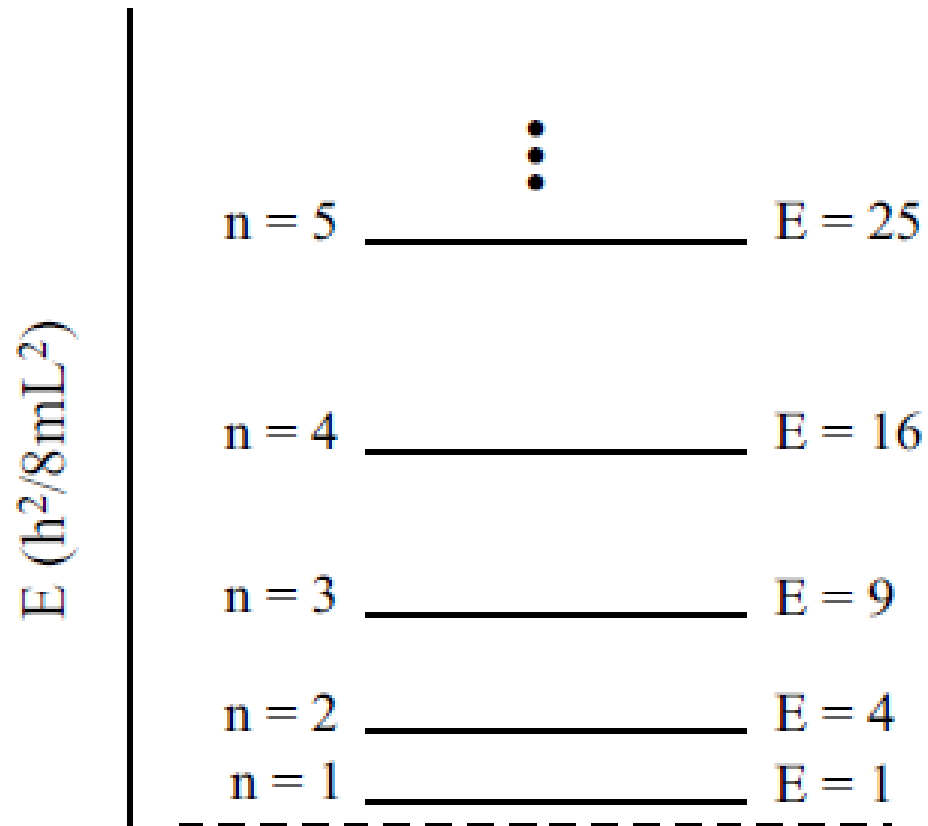
$$E = \frac{n^2h^2}{8mL^2}$$

A Discrete set of energy levels

There is a discrete set of energy levels for a given mass, m , and a given box length, L . As the quantum number n takes on values, 1, 2, 3, etc., the energies are

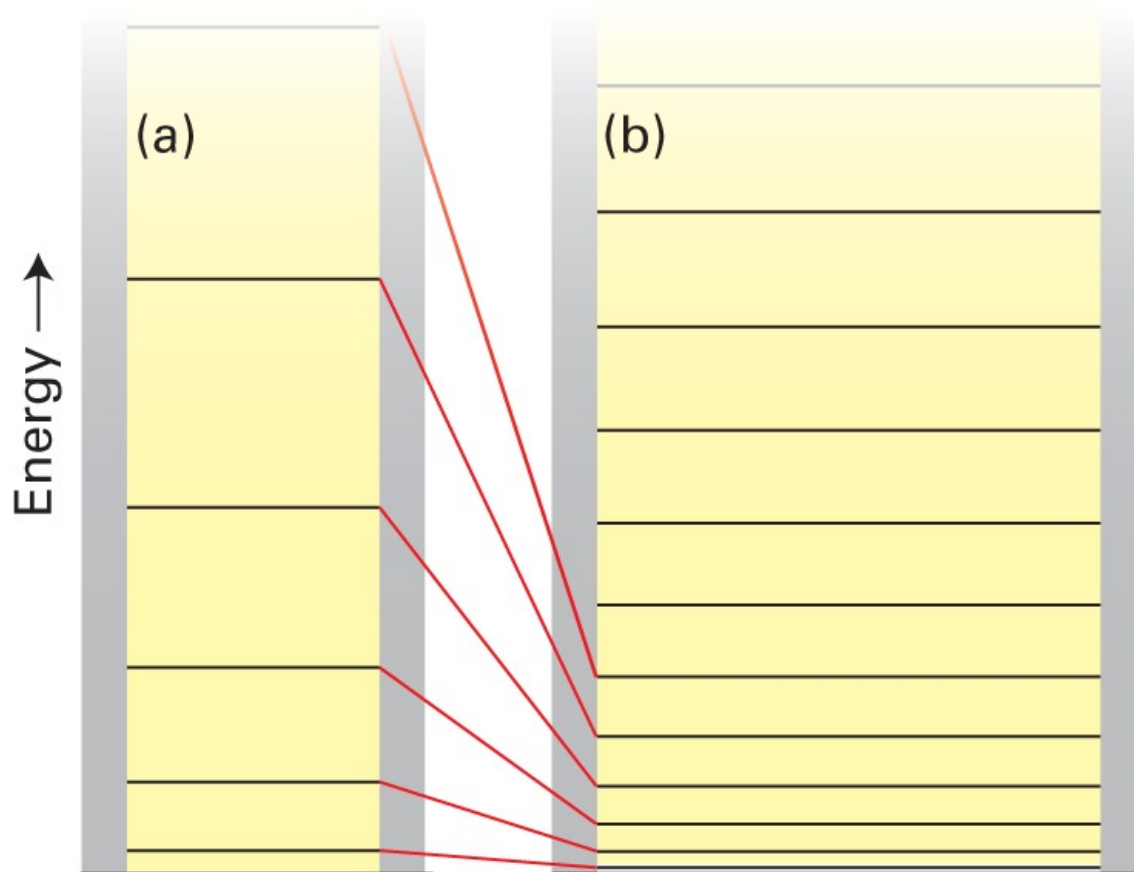
$$\frac{h^2}{8mL^2}, \frac{4h^2}{8mL^2}, \frac{9h^2}{8mL^2}, \text{ etc.}$$

Energy in units of $\frac{h^2}{8mL^2}$



$$\Delta E = E_{n+1} - E_n = \overbrace{(n+1)^2}^{n^2+2n+1} \frac{h^2}{8mL^2} - n^2 \frac{h^2}{8mL^2}$$

$$= (2n+1) \frac{h^2}{8mL^2}$$



Why are Cherries Red and Blueberries
blue ?

The Colour of Fruit

Let's put in numbers. $h = 6.6 \times 10^{-34}$ J-s. The electron mass, $m_e = 9.1 \times 10^{-31}$ kg. For the length of the box, let's take L to be that of a medium-sized molecule, that is, $L = 0.8 \times 10^{-9}$ m (0.8 nanometers, 0.8 nm). Then,

$$\Delta E = \frac{3(6.6 \times 10^{-34})^2}{8(9.1 \times 10^{-31})(0.8 \times 10^{-9})^2} = 2.8 \times 10^{-19} \text{ J.}$$

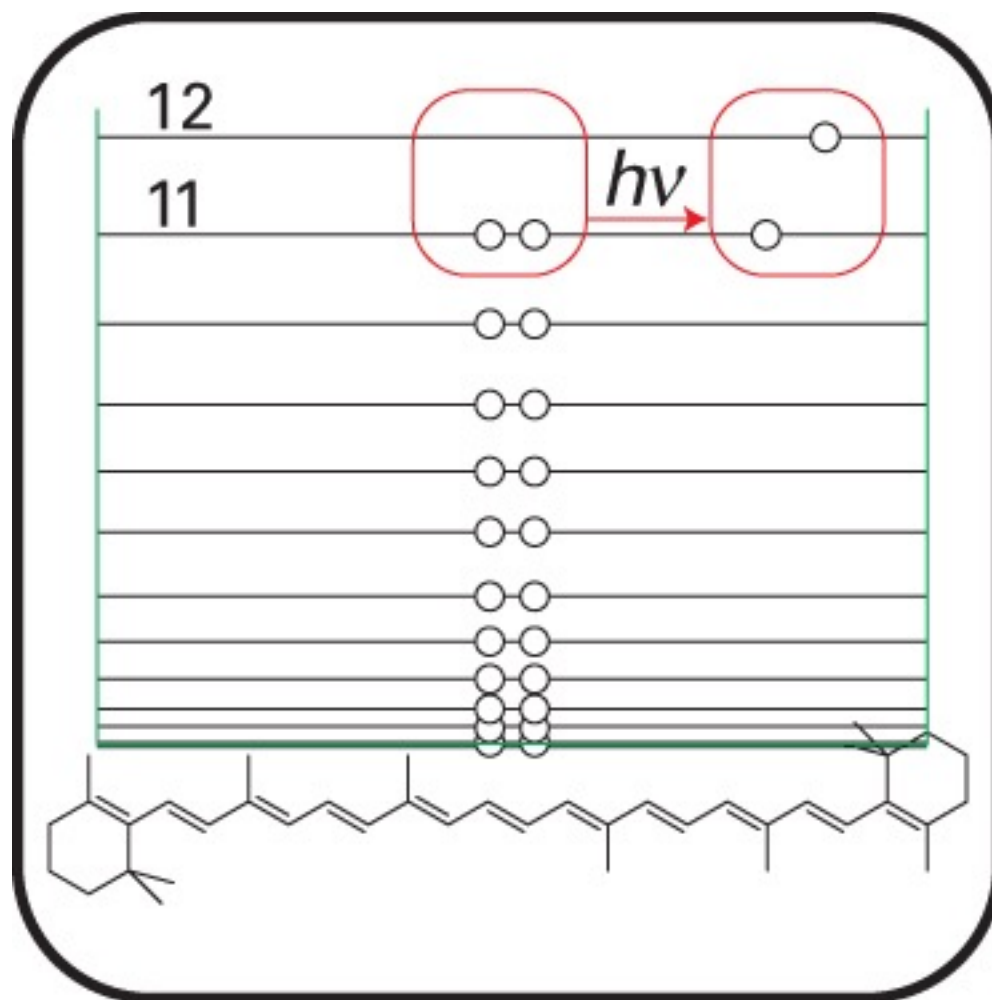
n=1 to n=2

This energy corresponds to

$$\lambda = 7.06 \times 10^{-7} \text{ m} = 706 \text{ nm} \longrightarrow \text{Deep Red Colour}$$

If $L=0.7$ nm, $\lambda=540$ nm **Green Colour**

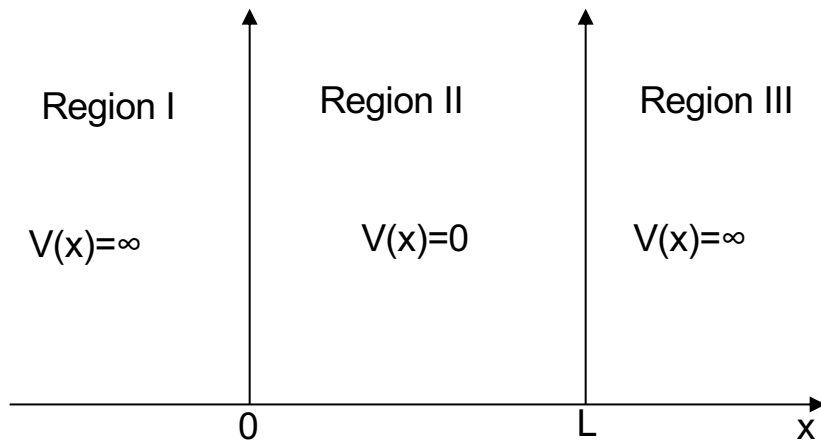
If $L=0.6$ nm, $\lambda=397$ nm **Blue Colour**



Particle in 1-dimensional box

- Infinite walls

Time Independent Schrödinger Equation



$$\underbrace{\frac{-\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2}}_{\text{KE}} + \underbrace{V(x)\psi}_{\text{PE}} = \underbrace{E\psi}_{\text{TE}}$$

Applying boundary conditions:

Region I and III:

$$\frac{-\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + \infty * \psi = E\psi$$

$$\Rightarrow |\psi|^2 = 0$$

$$\frac{d^2 \varphi(x)}{dx^2} = -\frac{2mE}{\hbar^2} \varphi(x)$$

Second derivative of a function equals a negative constant times the same function.

Functions with this property $\longrightarrow$ sin and cos.

$$\frac{d^2 \sin(ax)}{dx^2} = -a^2 \sin(ax)$$

$$\frac{d^2 \cos(ax)}{dx} = -a^2 \cos(ax)$$

Region II:

$$\frac{-\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = E\psi$$

$$\Rightarrow -\frac{d^2\psi(x)}{dx^2} = \frac{2m}{\hbar^2} E\psi$$

This is similar to the general differential equation:

$$-\frac{d^2\psi(x)}{dx^2} = k^2\psi$$

$$\psi = A \sin kx + B \cos kx$$

Applying boundary conditions:

$$\text{a) } x=0 \quad \psi=0$$

$$0 = A \sin 0k + B \cos 0k$$

$$0 = 0 + B * 1$$

$$\Rightarrow B = 0$$

$$\text{b) } x=L \quad \psi=0$$

$$0 = A \sin kL$$

$$\text{But } A \neq 0$$

$$kL = n\pi$$

Thus, wave function:

$$\psi_{II} = A \sin \frac{n\pi x}{L}$$

But what is 'A'?

Normalizing wave function:

$$\int_0^L (A \sin kx)^2 dx = 1$$

$$|A|^2 \left[\frac{x}{2} - \frac{\sin 2kx}{4k} \right]_0^L = 1$$

$$|A|^2 \left[\frac{L}{2} - \frac{\sin 2 \frac{n\pi}{L} L}{4 \frac{n\pi}{L}} \right] = 1$$

$$|A|^2 \left(\frac{L}{2} \right) = 1$$

$$|A| = \sqrt{\frac{2}{L}}$$

Thus normalized wave function is:

$$\psi_{II} = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$$

Calculating Energy Levels:

$$k^2 = \frac{2mE}{\hbar^2}$$

$$E = \frac{k^2 \hbar^2}{2m}$$

$$E = \frac{k^2 h^2}{2m 4\pi^2} \quad (\hbar = \frac{h}{2\pi})$$

$$E = \frac{n^2 \pi^2}{L^2} \frac{h^2}{2m 4\pi^2}$$

Thus Energy is:

$$E = \frac{n^2 h^2}{8mL^2}$$

Electronic material

Modelling an electronic material as a box allows us to ignore atoms.

Boxes can be of different dimensions:

Typical bulk material (Three dimension)

Quantum wells (2-dimension)

Quantum wire (1-dimension)

Quantum dot (0-dimension)

In quantum dots, electrons are confined in all three dimensions, in quantum wires, electrons are confined in only two dimensions and so on. So when we say that a given structure is 2-d, we mean that the electron is *unconfined* in 2 dimensions. In the unconfined directions, we will assume that the electron is described by a plane wave.

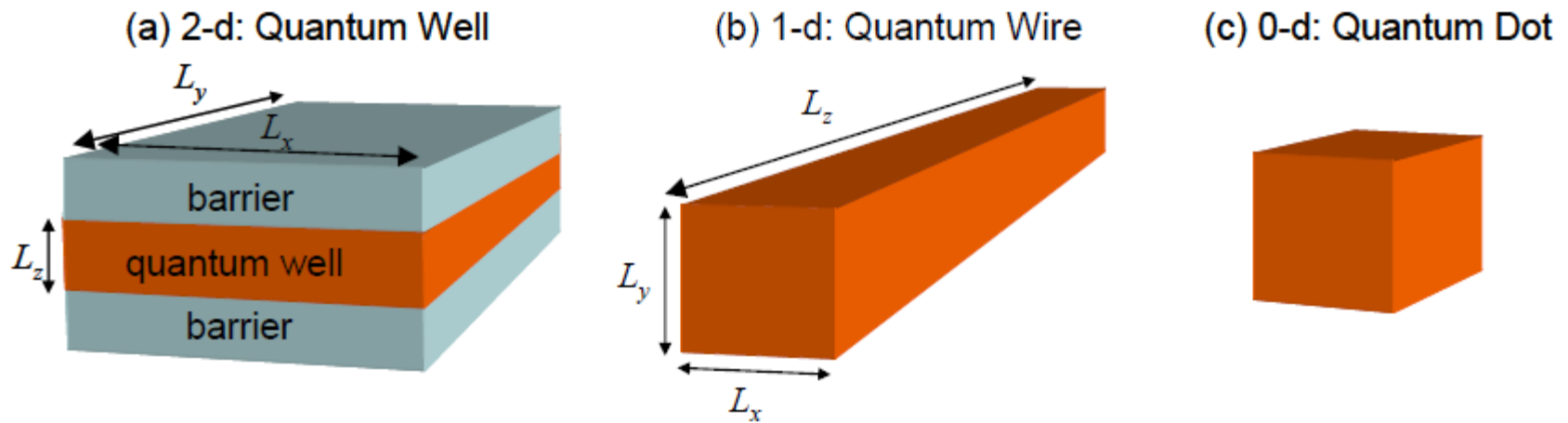


Fig. 2.11. (a) In quantum wells, electrons are confined only in one dimension. Quantum wells are usually implemented by burying the confining material within a barrier material. (b) Quantum wires confine electrons in two dimensions. The electron is not confined along the wire. (c) In a quantum dot, an electron is confined in three dimensions.

Quantum Well potential

$$V_x(x) = 0$$

$$V_y(y) = 0$$

$$V_z(z) = V_0 u(z - L) + V_0 u(-z)$$

$$E = E_x + E_y + E_z = \frac{\hbar^2 k_x^2}{2m} + \frac{\hbar^2 k_y^2}{2m} + \frac{n^2 \hbar^2 \pi^2}{2mL^2}$$

Quantum dot or Nanocrystals:

$$E_{n,m,l} = \frac{\pi^2 \hbar^2}{2m^*} \left(\frac{n^2}{L_z^2} + \frac{m^2}{L_y^2} + \frac{l^2}{L_x^2} \right), \psi = \phi(z) \phi(y) \phi(x) \text{-----} \quad (2)$$

Quantum wire:

$$E_{n,m}(k_x) = \frac{\pi^2 \hbar^2}{2m^*} \left(\frac{n^2}{L_z^2} + \frac{m^2}{L_y^2} \right) + \frac{\hbar^2 k_x^2}{2m^*}, \psi = \phi(z) \phi(y) \exp(ik_x x) \text{-----} \quad (3)$$

Quantum well:

$$E_n(k_x k_y) = \frac{\pi^2 \hbar^2 n^2}{2m^* L_z^2} + \frac{\hbar^2}{2m^*} (k_x^2 + k_y^2), \psi = \phi(z) \exp(ik_x x + ik_y y) \text{-----} \quad (4)$$

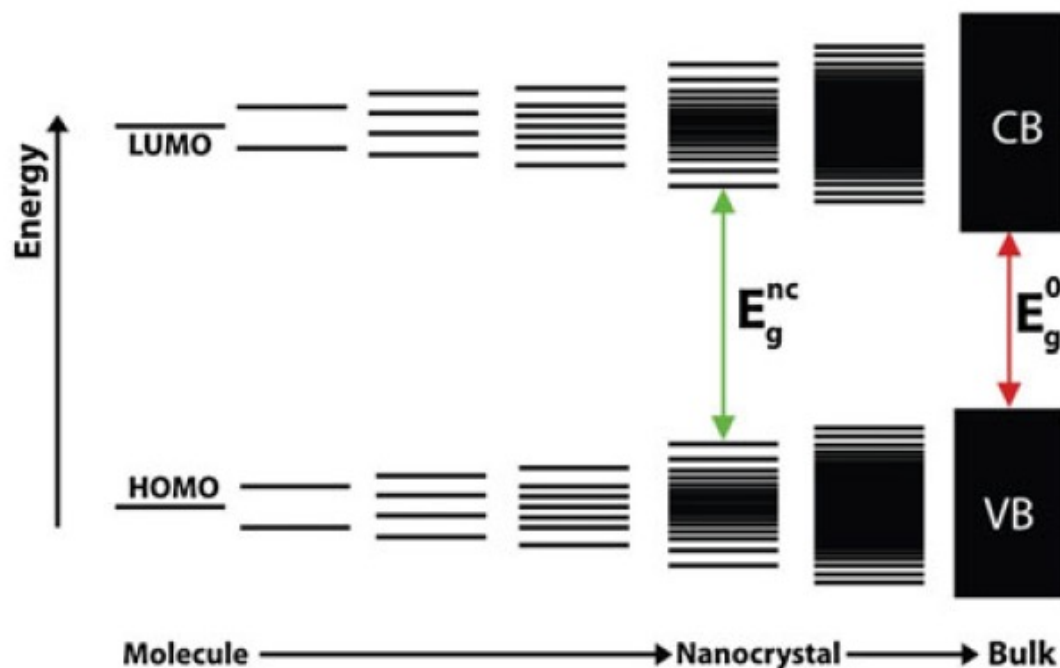
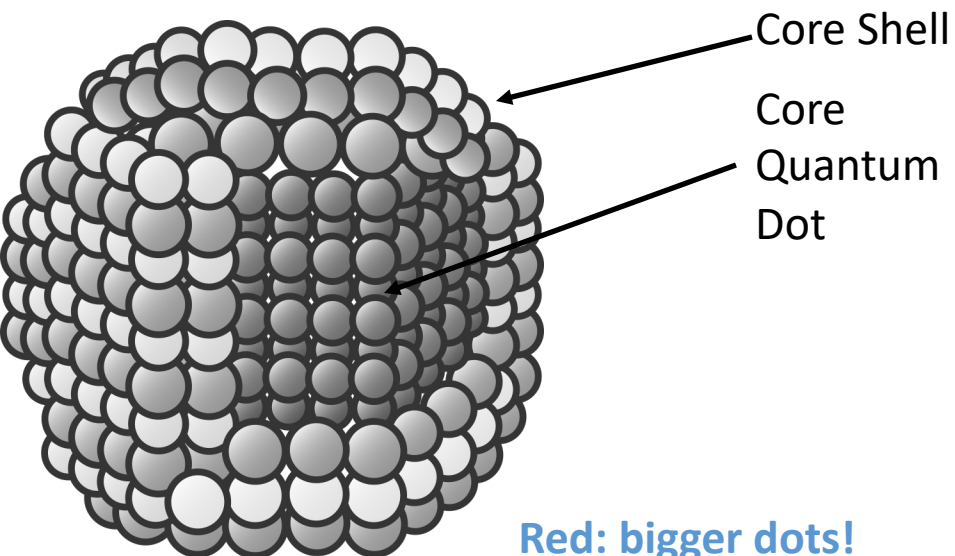


Fig. 2.12 Evolution of the energy level structure from a hypothetical diatomic molecule (*extreme left*) to a bulk semiconductor (*extreme right*). E_g^{nc} and E_g^0 indicate the energy gap between the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) for a nanocrystal and bulk, respectively (CB = conduction band, VB = valence band). Reprinted with permission from Ref. [45]

Semiconductor Nanoparticles

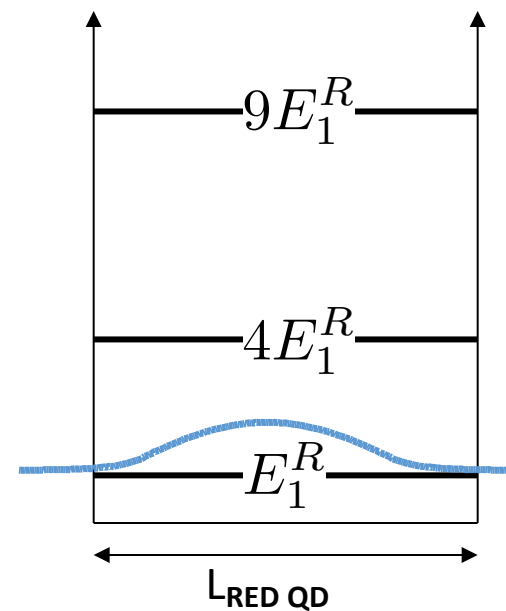
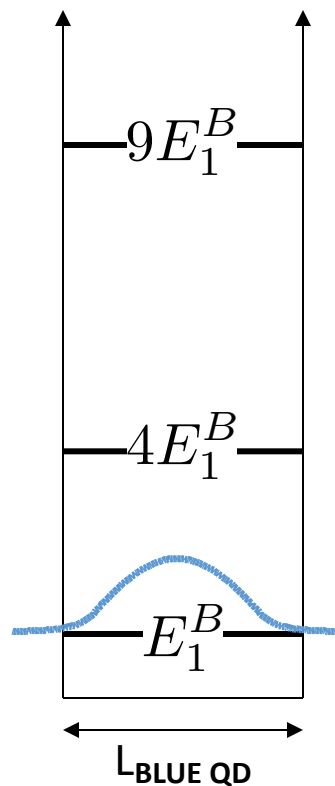
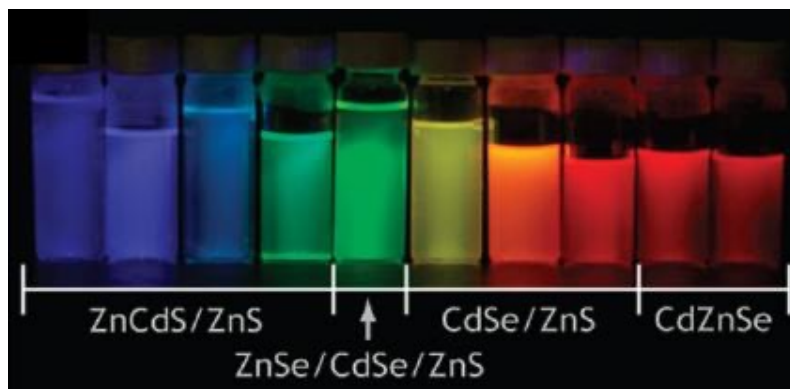
(aka: Quantum Dots)



Red: bigger dots!
Blue: smaller dots!

Determining QD energy
using the Schrödinger Equation

$$E_1 = n^2 E_1 \quad E_1 = \frac{\hbar^2 k_1^2}{2m} = \frac{\hbar^2 \pi^2}{2mL^2}$$



Hydrogen or Hydrogen type Atom (consists of a nucleus and just one electron):

Potential energy,

$$V(r) = \frac{-Ze^2}{4\pi\epsilon_0 r}$$

where, Ze = charge of the nucleus (Z is atomic number & e is charge of an electron)

ϵ_0 = permittivity of vacuum

r = distance from the nucleus to the electron

Hydrogen and hydrogen like atom

- **Hydrogen or Hydrogen type Atom**
(consists of a nucleus and just one electron)
- **Examples are H , He^+ , Li^{+2}**

Spherical coordinates

$$x = r \sin \theta \cos \phi$$

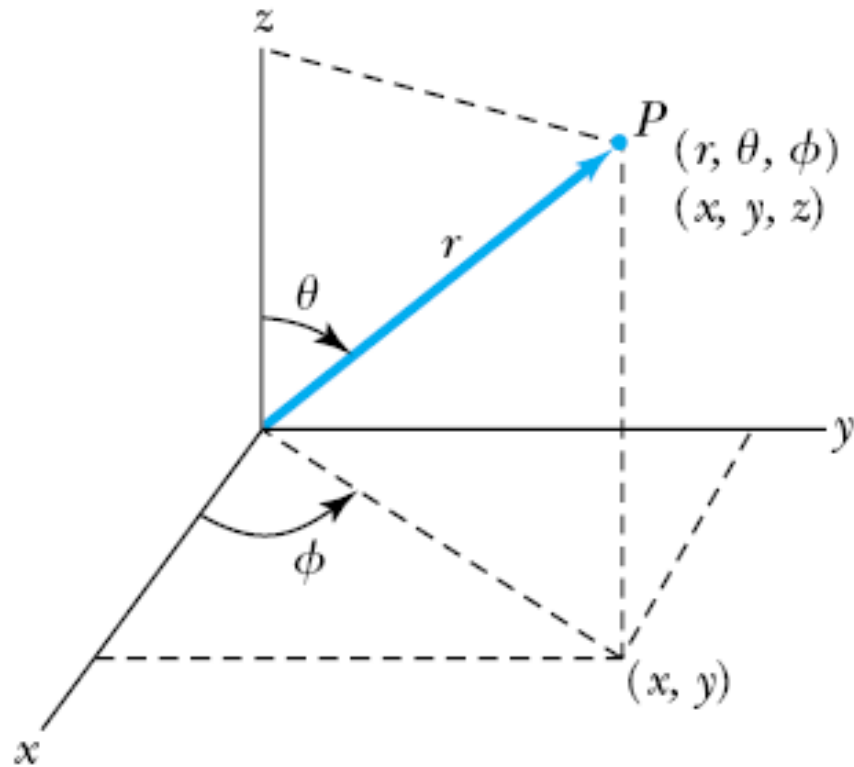
$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

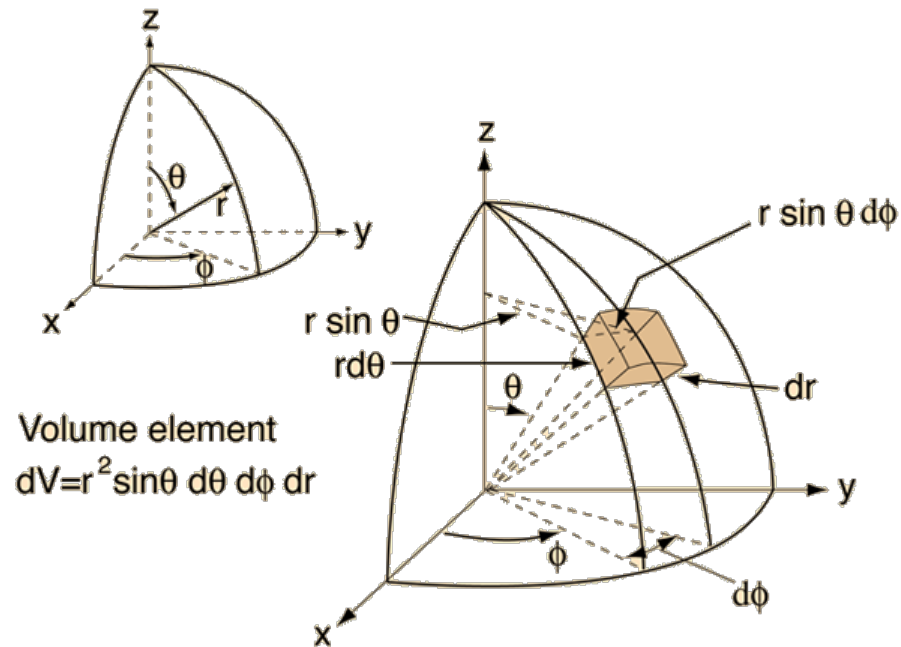
$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \cos^{-1} \frac{z}{r} \text{ (Polar angle)}$$

$$\phi = \tan^{-1} \frac{y}{x} \text{ (Azimuthal angle)}$$



Spherical polar coordinate



Hamiltonian in spherical polar coordinate

$$\nabla^2 = \frac{1}{r^2} \frac{d}{dr} r^2 \frac{d}{dr} + \frac{1}{r^2 \sin \theta} \frac{d}{d\theta} \sin \theta \frac{d}{d\theta} + \frac{1}{r^2 \sin^2 \theta} \frac{d^2}{d\phi^2}$$

$$\psi_{n,l,m_l}(r,\theta,\varphi) = \overbrace{R_{n,l}(r)}^{\text{Radial wavefunction}} \times \overbrace{Y_{l,m_l}(\theta,\varphi)}^{\text{Angular wavefunction}}$$

$$\left[-\left(\frac{\hbar^2}{2\mu}\right) \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{\hbar^2 l(l+1)}{2I} - \frac{Ze^2}{4\pi\epsilon_0 r} - E \right] R = 0$$

n	l	R_{nl}
1	0	$2a_o^{-3/2}e^{-r/a_o}$
2	0	$2^{-1/2}a_o^{-3/2}\left(1-r/2a_o\right)e^{-r/2a_o}$
2	1	$24^{-1/2}a_o^{-3/2}\left(r/a_o\right)e^{-r/2a_o}$
3	0	$\frac{2}{\sqrt{27}}\frac{1}{a_o^{3/2}}\left(1-\frac{2}{3}\frac{r}{a_o}+\frac{2}{27}\frac{r^2}{a_o^2}\right)e^{-r/3a_o}$
3	1	$\frac{8}{27\sqrt{6}}a_o^{-3/2}\left(1-\frac{1}{6}\frac{r}{a_o}\right)\left(\frac{r}{a_o}\right)e^{-r/3a_o}$
3	2	$\frac{4}{81\sqrt{30}}a_o^{-3/2}\left(\frac{r}{a_o}\right)^2e^{-r/3a_o}$

Hydrogen Atom

$$E_n = -\frac{\hbar^2}{2\mu} \gamma^2 = -\frac{\mu}{2\hbar^2 n^2} \frac{Z^2 e^4}{(4\pi\epsilon_0)^2} = -\frac{\mu e^4}{8\hbar^2 n^2} \frac{1}{(\epsilon_0)^2}$$

$$\Delta E = -\frac{\mu e^4}{8\hbar^2 (\epsilon_0)^2} \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$$

$$\nu = -\frac{\mu e^4}{8\hbar^3 (\epsilon_0)^2} \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$$

$$\nu = -R_H \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right) \quad R_H = 1.096776 * 10^7 m^{-1}$$

Recognizing that this equation can be separated into θ and ϕ parts:

$$\frac{\sin \theta}{\Theta(\theta)} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \Theta(\theta)}{\partial \theta} \right) + \beta \sin^2 \theta = m^2$$
$$\frac{1}{\Phi(\phi)} \frac{\partial^2 \Phi(\phi)}{\partial \phi^2} = -m^2$$

$$Y_0^0(\theta, \phi) = \frac{1}{(4\pi)^{1/2}}$$

$$Y_1^1(\theta, \phi) = -\left(\frac{3}{8\pi}\right)^{\frac{1}{2}} \sin \theta e^{i\phi}$$

$$Y_2^0(\theta, \phi) = \left(\frac{5}{16\pi}\right)^{\frac{1}{2}} (3 \cos^2 \theta - 1)$$

$$Y_2^{-1}(\theta, \phi) = \left(\frac{15}{8\pi}\right)^{\frac{1}{2}} \sin \theta \cos \theta e^{-i\phi}$$

$$Y_2^{-2}(\theta, \phi) = \left(\frac{15}{32\pi}\right)^{\frac{1}{2}} \sin^2 \theta e^{-2i\phi}$$

$$Y_1^0(\theta, \phi) = \left(\frac{3}{4\pi}\right)^{\frac{1}{2}} \cos \theta$$

$$Y_1^{-1}(\theta, \phi) = \left(\frac{3}{8\pi}\right)^{\frac{1}{2}} \sin \theta e^{-i\phi}$$

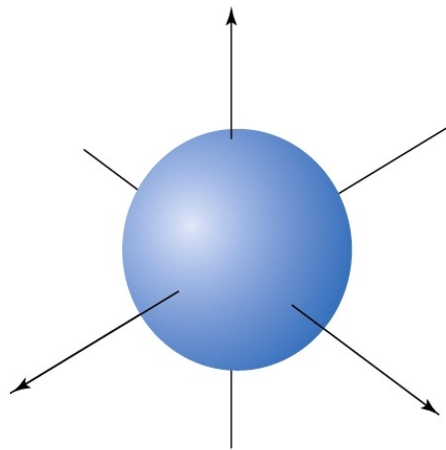
$$Y_2^1(\theta, \phi) = -\left(\frac{15}{8\pi}\right)^{\frac{1}{2}} \sin \theta \cos \theta e^{i\phi}$$

$$Y_2^2(\theta, \phi) = \left(\frac{15}{32\pi}\right)^{\frac{1}{2}} \sin^2 \theta e^{2i\phi}$$

First spherical harmonics

Spherical Harmonic, $Y_{0,0}$

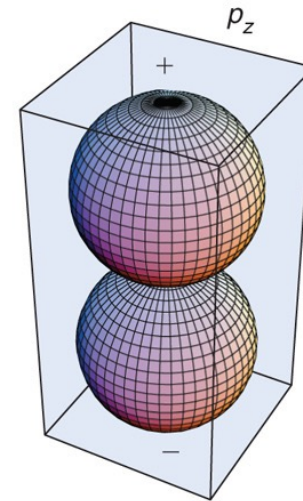
$$|Y_0^0|^2 = \text{const} = \text{distance of surface from the origin}$$



$$l = 1, m = 0$$

$$\left| Y_1^0 \right|^2 = \text{const} \times \cos^2 \theta = \text{distance of surface from the origin}$$

θ	$\cos^2 \theta$
0	1
30	3/4
45	1/2
60	1/4
90	0



$l=2, m=0$

$$Y_{2,0}(\theta, \phi) = \frac{\sqrt{5}}{4\sqrt{\pi}} (3\cos^2(\theta) - 1)$$

θ	$\cos^2\theta$	$3\cos^2\theta - 1$
0	1	2
30	3/4	$(9/4 - 1) = 5/4$
45	1/2	$(3/2 - 1) = 1/2$
60	1/4	$(3/4 - 1) = -1/4$
90	0	-1

